

→ Example 2:-

* Consider the function:

$$\text{Max-list } [] = 0$$

$$\text{Max-list } (x :: xs) = \max x (\text{Max-list } xs)$$

* Also consider

$$\text{List-set } [] = \emptyset$$

$$\text{List-set } (x :: xs) = \text{insert } x (\text{List-set } xs)$$

Note: "insert" adds an element to a given list

$$x \in \text{List-set } xs \Rightarrow x \leq \text{Max-list } xs$$

proof by induction on xs

Note: The property is "assumption $\Rightarrow$ conclusion", not just the conclusion!!!

Base case

assumptions: $x \in \text{List-set } []$

goal is: $x \leq \text{Max-list } []$

1. $\text{Max-list } [] = 0$ by Max-list.simp

2. $x \in \emptyset$ from base case assumption, & subst List-set.simp

3. False using $\text{Lem3 } [2, \text{Lem1}]$

4. $x \leq \text{Max-list } []$ using $\text{Lem2 } [0 \text{ of } 3]$

lem1: $x \notin \emptyset$

lem2: $\text{false} \Rightarrow P$

lem3: $P \Rightarrow \neg P \Rightarrow \text{false}$

→ * step case

assumptions:

$$\text{I.H. } x \in \text{List-set } ys \Rightarrow x \leq \text{Max-list } ys$$

$$\text{prem } x \in \text{List-set } (y :: ys)$$

$$\text{goal: } x \leq \text{Max-list } (y :: ys)$$

proof by cases on $x = y$

case 1: $x = y$

1. $y \leq \text{Max-list } (y :: ys)$ from Lem1

2. $x \leq \text{Max-list } (y :: ys)$ by subst. case assumption

case 2: $x \neq y$

1. $x \in \text{List-set } ys$ from case assumption & Lem2

2. $x \leq \text{Max-list } ys$ from I.H. [OF 1]

3. $\text{Max-list } ys \leq \text{Max-list } (y :: ys)$ using Lem3

4. $x \leq \text{Max-list } (y :: ys)$ using $\text{Lem4 } [0 \text{ of } 2, 3]$

lem4: $a \leq b \Rightarrow b \leq c \Rightarrow a \leq c$